

Verify Before You Execute: A Hybrid Symbolic–Learned Pre-Execution Verifier for LLM-Generated Plans

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Abstract

LLM agents increasingly emit multi-step plans—action sequences, tool-call chains—that downstream systems execute directly. Executing a flawed plan is costly, yet the dominant safeguards are either an LLM judging its own kind or no check at all. We present a pre-execution verifier that translates a free-text plan into a typed action schema, *soundly* checks consistency, goal completeness, and resource feasibility by forward simulation, and fuses the symbolic verdict with a small calibrated *trust model* that estimates whether the NL→schema translation itself was faithful. Hard symbolic failures always reject; the trust score ranks residual translation risk. On three domains (Blocksworld, Logistics, and a tool-use environment that reused the pipeline with zero code changes), against oracle labels validated by two independent simulators agreeing on 360/360 records, the hybrid attains $F1=0.949$ at perfect recall and—most importantly—raises downstream execute-or-reject task success from 61.1% (no verifier) to 93.1%, beating an LLM-judge gate (87.5%) because its symbolic explanations make replanning feedback actionable. At the plan lengths typical of prior LLM-planning benchmarks (3–8 steps), properly-budgeted LLM judges are in fact *perfect* detectors, matching the hybrid. We therefore run a second experiment scaling plan length to 10/20/40/80 steps and find the picture reverses: the hybrid holds $F1=1.000$ at every horizon while judge recall degrades monotonically (0.94→0.84 zero-shot, 0.92→0.88 chain-of-thought), and in a live downstream execute-or-reject test at 10–20 steps the judge-gated arm lets 4–5% of flawed plans execute while the hybrid remains at exactly 0% at every horizon tested—the one guarantee sound simulation gives that judging cannot. We also report two further negative-space findings: a constrained output format makes the NL→schema translation lossless, eliminating the trust model’s target class in that regime; and a naive token budget silently turned a fail-open judge baseline into “no verifier” on 100% of records, a deployment failure mode worth naming on its own. Code, data generators, and all results are reproducible from a single make target on a laptop.

1 Introduction

A growing class of systems lets a large language model (LLM) propose a multi-step plan—a sequence of robot actions, API calls, or tool invocations—and then executes that plan with little or no intermediate checking. When the plan is

flawed, the cost is paid at execution time: side-effecting API calls fire, budgets are spent, and physical or irreversible actions occur before the flaw surfaces. Benchmarks consistently find that LLM plans are unreliable in exactly the ways that matter: preconditions are violated, goal conjuncts are silently dropped, and resource limits are exceeded (Valmeekam et al. 2023; Kambhampati et al. 2024).

The natural safeguards occupy two unsatisfying extremes. Purely symbolic validation in the style of VAL (Howey, Long, and Fox 2004) is sound but assumes the plan already exists in a formal language—while LLM plans arrive as free text, and the lossy translation from text to formalism becomes an unmodeled failure source. Purely learned safeguards—asking an LLM judge “is this plan valid?” (Zheng et al. 2023)—inherit the fallibility they are meant to check, and recent evidence suggests LLMs are poor self-verifiers (Huang et al. 2024; Stechly, Valmeekam, and Kambhampati 2025).

We study the middle: a **hybrid pre-execution verifier** for LLM plans. An LLM extractor translates each free-text step into a typed action schema; a sound symbolic checker forward-simulates the extracted plan and verdicts three independent dimensions (action consistency, goal completeness, resource feasibility); and a small calibrated *trust model*—trained on extraction confidence, k -resample self-consistency agreement, and plan-shape features—estimates the probability that the symbolic verdict is *faithful* to the plan as written, i.e. that the translation did not silently change what is being checked. The fusion rule is asymmetric by design: a hard symbolic failure always rejects, regardless of trust; among symbolic passes, low trust flags residual risk.

Contributions. (1) A pipeline architecture and open implementation for sound pre-execution verification of free-text LLM plans, with a learned estimate of translation faithfulness as a first-class signal. (2) A synthetic-data methodology with an *independent* oracle: two separately implemented simulators (labeler and verifier) whose agreement on every record (360/360; 1,440/1,440 per-dimension comparisons) anchors all reported numbers. (3) A generality test: a tool-use domain (auth scopes, prerequisite calls, quotas, spend budgets) was expressed in the same DSL and ran through the entire pipeline with *zero* code changes. (4) A downstream execute-or-reject experiment showing the hybrid’s real edge at short horizons is not detection accuracy but *replan feed-*

back quality: symbolic explanations convert rejections into successful second attempts more often than judge critiques (5 vs. 9 dead-ends). (5) A plan-length scaling experiment (10/20/40/80 steps) showing that detection accuracy parity is a short-horizon artifact: judge recall degrades monotonically with length while sound simulation does not, and by 10–20 steps the judge-gated downstream arm already lets flawed plans execute at a nonzero rate that the hybrid never does. (6) Several honestly-reported negative and boundary results that reframe the design space for verifier research, including a deployment failure mode in which a fail-open LLM judge silently degrades to no verification at all.

2 Related Work

LLM planning and its critics. PlanBench and successors document systematic planning failures in LLMs (Valmeekam et al. 2023). The LLM-Modulo position argues LLMs should generate while external sound modules verify (Kambhampati et al. 2024); LLM+P (Liu et al. 2023) translates problems into PDDL and delegates to a classical planner. We keep the LLM as planner and verify its output, targeting settings (e.g. tool use) where a complete classical model exists only implicitly.

Plan validation. VAL (Howey, Long, and Fox 2004) validates PDDL2.1 (Fox and Long 2003) plans against a formal domain. Our symbolic core is a deliberately small reimplementation of this idea over a typed DSL with per-step resource accounting; the novelty is not the checker but its coupling to a learned model of *whether the checker saw the right plan*.

LLM self-checking and repair. LLM-as-judge (Zheng et al. 2023), Self-Refine (Madaan et al. 2023), and Reflexion (Shinn et al. 2023) use LLMs to critique or repair their own outputs; Huang et al. (2024) and Stechly, Valmeekam, and Kambhampati (2025) find intrinsic self-correction weak on reasoning and planning. Our baselines instantiate both judge variants and a Reflexion-style repairer on identical inputs, and our downstream experiment measures what each safeguard is worth at execution time.

Calibration and self-consistency. We use k -resample agreement in the spirit of self-consistency (Wang et al. 2023) as a *feature* rather than a decoding rule, Platt scaling (Platt 1999) for calibration, and expected calibration error (Naeini, Cooper, and Hauskrecht 2015; Guo et al. 2017) for evaluation.

3 Method

3.1 Plan and Domain Representation

A domain is a set of typed predicate definitions, action schemas (preconditions, add/delete effects), and *resource dimensions*—named scalars with an initial value, a floor, and a cap, adjusted by per-action deltas. A problem grounds a domain with typed objects, an initial state, and a conjunctive goal. Validation is structural: an action schema may only reference declared predicates and resources with matching

arity and types. This DSL is intentionally small—no disjunctive preconditions, conditional effects, or typed return values—but Section 5.7 shows it stretches to tool-use constraints without modification.

3.2 Symbolic Verifier

Given a sequence of ground actions, the verifier forward-simulates from the initial state and emits a three-part verdict with a machine-readable explanation: **consistency** (every step’s action exists, is well-typed, and its preconditions hold when executed in order), **goal completeness** (every goal conjunct holds in the final state), and **resource feasibility** (every prefix of the running sum respects every dimension’s floor and cap). Progression is *lenient*: a violated precondition is recorded but effects still apply, so one early slip cannot mask an independent later flaw—each dimension verdicts independently. Upstream parse or extraction failures are injected as consistency violations, since an executor could not run those steps. `overall_valid` is the conjunction of the three checks.

3.3 NL→Schema Extraction with Self-Consistency

Free-text steps are mapped to ground actions by an LLM extractor using the API’s structured-output mode, validated against the domain (action name, arity, object types). On validation failure the extractor retries once with the error as feedback, then falls back to a null extraction with confidence 0—it never crashes, it degrades. Each step is resampled $k=3$ times at temperature 0.7; the modal extraction is used, and two agreement statistics (exact-match fraction and action-type-match fraction) are retained. These, with the extractor’s self-reported confidence, become features rather than being discarded.

3.4 Trust Model and Fusion

The trust model answers: *given how the extraction went, how likely is the symbolic verdict faithful to the plan as written?* Its label is agreement between the production verdict (LLM extraction path) and a reference verdict computed by a deterministic rule-based parse of the original plan text. Features (17) span extraction confidence (mean/min), k -resample agreement, plan-shape statistics, the fraction of failed-extraction steps, a TF-IDF character- n -gram similarity between the preconditions the extractor *mentioned* and those the schema *requires*, the symbolic verdict bits and violation counts, and a domain indicator. The classifier is a class-balanced logistic regression, Platt-scaled on a held-out validation split; all splits are by *problem*, so no problem’s records straddle train/test.

Fusion is a two-clause rule with a tested invariant:

$$\text{reject} \iff \neg \text{symbolic_valid} \vee \text{trust} < \tau,$$

where the first clause is unconditional—no trust score can rescue a hard symbolic failure—and τ is picked on validation. $\tau=0$ recovers symbolic-only exactly.

4 Experimental Setup

4.1 Domains, Problems, and Gold Plans

Blocksworld (4 operators + a crane-energy resource), **Logistics** (trucks/planes over two cities, fuel + budget), and **Tools** (Section 5.7). A seeded generator produces problems whose BFS-optimal gold plans span 3–8 steps; determinism is enforced cross-process (a `PYTHONHASHSEED` ordering bug found by a subprocess test motivated sorting all set-derived output). Resource caps are tightened to $\lceil 1.25 \times \rceil$ the gold plan’s consumption so that wasteful plans can actually overrun.

4.2 Long-Horizon Problems

Prior LLM-planning benchmarks concentrate on short plans, and our own main dataset (3–8 steps) is no exception—raising the question of whether any detector comparison run at that length generalizes. To test this directly we build a second, constructive generator per domain (BFS gold-planning is infeasible past ~ 10 steps) that scales problem size (blocks, packages, trips) to hit nominal horizons of 10, 20, 40, and 80 reference-plan steps: Blocksworld tears every tower down to the table and rebuilds the goal arrangement bottom-up; Logistics routes each package sequentially through its truck/airport/plane legs; Tools authenticates each needed scope once and executes every trip’s milestone prerequisite chain in order. Every constructive plan is independently strict-simulated at generation time and re-checked by the oracle labeler, so a bug in a constructive planner cannot silently ship an invalid reference plan; the one honest caveat is that these plans are valid but *not optimal* (Blocksworld especially), so “horizon H ” denotes a band on the reference length, not an optimality claim. To bound API cost, the k -resampled extraction path runs on all 40 records per horizon at $H \in \{10, 20\}$ but on a stratified 12-record subset at $H \in \{40, 80\}$; judges and self-repair run on all records at every horizon. Token budgets were raised throughout (generation/paraphrase to 4,096, judges/self-repair to 8,192 tokens) so long plans are not silently truncated the way Section 5.5 describes.

4.3 Plan Generation with Failure Injection

For each problem, `claude-haiku-4-5` (temperature 1.0) writes a plan under four prompt conditions: *baseline* (honest prompt), *goal-omission* (one goal conjunct silently dropped from the prompt; labels use the full goal), *resource-blind* (limits stripped), and *distractor* (plausible invalid actions offered). A pilot calibration confirmed the conditions amplify naturally occurring failure modes rather than inject alien ones: 43–47% of *baseline* plans are already flawed, and goal-omission roughly quadruples goal-incompleteness. Dataset: 3 domains $\times$ 30 problems $\times$ 4 conditions = 360 records.

4.4 Independent Oracle and Soundness Evidence

Ground-truth labels come from an oracle labeler that is a *separate implementation* of plan simulation from the verifier (shared design semantics, disjoint code). Agreement between the two on rule-parsed plans is therefore evidence, not tautology: across all 360 records they agree on overall validity

System	P	R	F1
Hybrid (ours, τ picked on val)	0.903	1.000	0.949
Symbolic-only	0.903	1.000	0.949
Learned-only (no symbolic feats)	0.618	0.750	0.677
LLM judge, zero-shot	1.000	1.000	1.000
LLM judge, CoT	1.000	1.000	1.000
Self-repair (change detection)	0.966	1.000	0.982

Table 1: Detection on the 72-record pooled test split (positive class = flawed). Per-flaw recall for the hybrid: 20/20 inconsistency, 10/10 goal-incompleteness, 5/5 resource-infeasibility.

360/360 and on all 1,440 per-dimension comparisons, with every flaw type additionally covered by hand-verified adversarial fixtures. The evaluation CLI recomputes this cross-check on every run and fails loudly on any disagreement.

4.5 The Prose Regime

Our first finding is methodological: with a constrained output format (Step N : `action(args)`), the LLM extraction path’s verdict agreed with the rule-based reference on **240/240** records—the NL $\rightarrow$ schema translation was loss-less, the trust label had one class, and the learned layer had nothing to model. This is a genuine scoping result: *where plans can be forced into a rigid format, a rule parser plus a sound checker suffices*. To study the setting the trust model exists for, the production pipeline verifies an LLM-generated *prose paraphrase* of each plan (one sentence per step, no rigid syntax), while labels stay anchored to the original constrained text. In the prose regime 13/360 records (3.6%) get unfaithful verdicts—sparse, but a real target class.

4.6 Baselines

On identical prose inputs: **LLM judge** (zero-shot and chain-of-thought, verdict parsed from a required `VERDICT`: line, fail-open on unparseable output, mirroring naive deployment); **symbolic-only** ($\tau=0$); **learned-only** (the trust feature vector with all five symbolic-verdict features removed, trained to predict validity directly—the no-grounding ablation); and **self-repair** (Reflexion-style single-pass critique+rewrite (Shinn et al. 2023), with change-detection standing in for a verdict).

4.7 Metrics

Positive class = *flawed plan*; rejecting is a positive prediction. We report P/R/F1 overall, per domain, and per flaw type; ECE with reliability diagrams for the trust model; API-call and local-latency cost proxies; and the downstream experiment of Section 5.4. Test split: 72 records (problem-level, 216/72/72).

5 Results

5.1 Detection

Table 1 contains a result we report at full prominence: **both LLM judges are perfect on this test split**. Plans of 3–8 steps are short enough for the judge model to simulate flawlessly when given an adequate token budget (see Section 5.5). At this horizon, “hybrid beats LLM-judge on detection” is *not*

Horizon	Hybrid F1	Judge-zs F1 (R)	Judge-CoT F1 (R)
5 (ref.)	0.949	1.000 (1.00)	1.000 (1.00)
10	1.000	0.967 (0.94)	0.960 (0.92)
20	1.000	0.950 (0.91)	0.976 (0.95)
40	1.000	0.937 (0.88)	0.962 (0.93)
80	1.000	0.914 (0.84)	0.938 (0.88)

Table 2: F1 (and recall in parentheses for the judges) vs. nominal plan horizon, pooled over domains. The hybrid is perfect at every horizon ≥ 10 ; judge recall degrades monotonically as plans lengthen.

supported by Table 1 alone—but Section 5.2 shows this parity is a short-horizon artifact that does not survive longer plans. The learned-only ablation (F1 0.677) confirms the symbolic grounding carries the detection signal.

The hybrid’s only errors are 3 false rejects, all on the Tools domain (precision there 0.833, perfect elsewhere), all with one signature: the extractor mangles hyphenated arguments (flights-scope $\rightarrow$ flights), producing unknown-object violations on genuinely valid plans.

5.2 Detection Parity Is a Short-Horizon Artifact

We repeat detection at nominal horizons of 10, 20, 40, and 80 steps (Section 4.2). Table 2 settles the question Section 5.3’s parity result left open: the hybrid (identical to symbolic-only at every horizon tested, since $\tau=0$ throughout) holds F1=1.000 and recall=1.000 at every horizon from 10 to 80 steps—expected, since forward simulation’s cost is linear in plan length and its correctness does not degrade with length. Both judges degrade monotonically in *recall* as horizon grows (zero-shot 1.00 $\rightarrow$ 0.94 $\rightarrow$ 0.91 $\rightarrow$ 0.88 $\rightarrow$ 0.84; CoT 1.00 $\rightarrow$ 0.92 $\rightarrow$ 0.95 $\rightarrow$ 0.93 $\rightarrow$ 0.88, noisier but the same downward drift, Figure 1)—they increasingly fail to *catch* flawed plans, not fail to parse: judge mean output tokens rise with horizon (zero-shot 634 $\rightarrow$ 854, CoT 765 $\rightarrow$ 1,192 tokens, Figure 2) and 0/480 judge calls were unparseable at any horizon, so this is a genuine simulation-accuracy failure, not a formatting artifact. The flaw rate among LLM-generated plans also rises sharply with horizon (38.9% at $H=5 \rightarrow$ 65% at $H=10 \rightarrow$ 87.5% at $H=20 \rightarrow$ 100% at $H=80$)—longer LLM-generated plans are both harder to verify *and* more likely to be flawed, compounding the value of a sound checker exactly where judges get worse.

The soundness gap this implies is not merely theoretical: we re-ran the downstream execute-or-reject experiment (Section 5.4) at $H \in \{10, 20\}$ (120 records each; $H=40, 80$ were not affordable to run downstream given the recall trend already visible at $H \leq 20$). At $H=5$ both gates achieved 0% executed-flawed. At $H=10$ the judge-gated arm lets 5% of records execute a flawed plan despite the gate; at $H=20$, 4.2%. The hybrid-gated arm remains at **exactly 0%** executed-flawed at both horizons—the one guarantee sound simulation gives that judging cannot, and the guarantee actually degrades under exactly the conditions (longer, more complex, more failure-prone plans) where a verifier is needed most. Task success falls for every arm as horizon grows (the underlying planner LLM struggles at 10–20 steps, indepen-

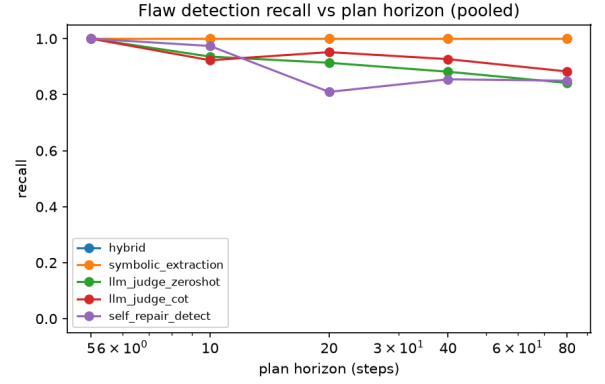


Figure 1: Flaw-detection recall vs. plan horizon, pooled. The hybrid stays at 1.0; both judges drift downward as plans lengthen.

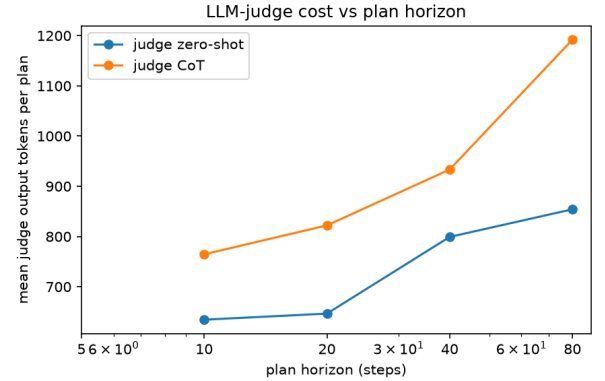


Figure 2: Mean judge output tokens vs. plan horizon. Judges spend more compute on longer plans yet still miss more flaws (Table 2).

dent of verification), but the hybrid still leads at both horizons (0.667 vs. 0.525 at $H=10$; 0.383 vs. 0.283 at $H=20$). We flag two limits on this result honestly: the long-horizon extraction cells are 12-record stratified subsets with correspondingly wide error bars, and the constructive gold plans are valid but not optimal (Section 4.2).

5.3 The Trust Signal Points the Right Way—but the Fusion Rule Cannot Use It

The validation-picked threshold is $\tau=0$: the trust layer never changes a decision at dev scale, and the test split contains zero symbolically-clean-but-flawed records for it to catch (41 symbolic passes, all genuinely valid). Yet the signal itself is demonstrably real: the three false-rejected mistranslations receive the trust model’s *lowest* scores in the entire test split (0.74–0.82, vs. >0.9 elsewhere; classifier accuracy 0.958, ECE 0.067, Figure 3). The model detects unfaithful translation exactly as designed—but our fusion rule, like prior gate-then-execute designs, only lets trust *add* rejections. A bidirectional rule (low-trust symbolic reject $\rightarrow$ re-extract, or

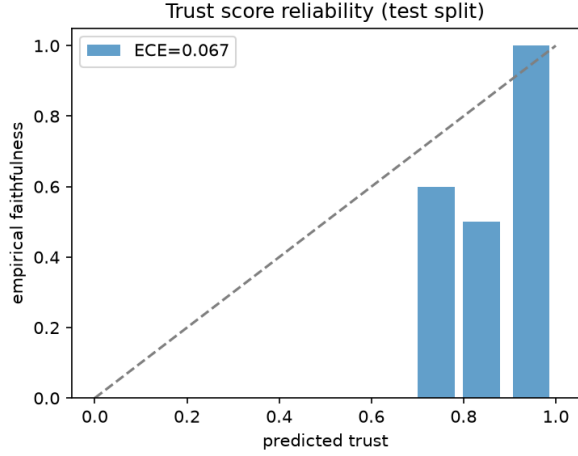


Figure 3: Trust-model reliability diagram (test split). Mass concentrates in the top bin at accuracy 1.0; the sparse low-confidence bins are exactly the mistranslated records of Section 5.3.

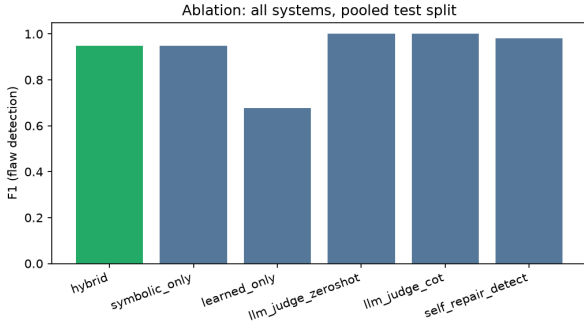


Figure 4: F1 by system and domain. The learned-only bar shows the cost of removing symbolic grounding; judge bars are at ceiling at this plan length.

fall back to the rule parser) would have repaired all three errors and is the concrete architectural lesson of this study.

5.4 Downstream: Execute or Reject

Table 3 is where the architecture earns its keep at this horizon. Both gates eliminate flawed execution entirely here, but the hybrid converts more rejections into successful second attempts: its replan feedback is the symbolic explanation itself (“step 3: unknown object . . .”, “goal conjunct on (b3, b2) unmet”), which is more actionable than the judge’s free-text critique—5 vs. 9 dead-ends, 93.1% vs. 87.5% final task success. At $H=5$: detection parity, feedback superiority. Section 5.2 reruns this exact experiment at longer horizons and finds the judge’s “eliminates flawed execution entirely” property itself breaks down.

Arm	Success	Flawed exec.	Replans	Dead ends
No verifier	0.611	0.389	0	0
Hybrid-gated	0.931	0.000	31	5
CoT-judge-gated	0.875	0.000	28	9

Table 3: Execute-or-reject with one feedback-driven replan (72 test records). Success = an executed plan that is actually valid.

5.5 A Deployment Failure Mode: the Silently Absent Judge

Our first zero-shot judge used a 64-token budget, reasoning that a one-line verdict needs no more. The judge model ignored the “respond with exactly one line” instruction, simulated the plan step by step, and was truncated before ever emitting its verdict—on **360/360 records**. Because unparseable output fails open (as a naive deployment would), the judge scored $P=R=0.0$ while appearing to run normally: the system silently became *no verifier at all*. At 512 tokens, 41% of outputs still truncated; only at 2,048 (the CoT budget) did parse failures reach zero—whereupon the judge jumped from 0.0 to a perfect 1.0. We fixed the baseline and report the fixed numbers, but the episode is itself a finding: fail-open verdict parsing plus format non-compliance is a silent total-failure mode, and the *zs-vs-CoT* contrast in any judge comparison is confounded by budget unless equalized.

5.6 Self-Repair and Cost

The Reflexion-style repairer is a surprisingly strong *fixer*—23/28 flawed plans repaired, 0/44 valid plans broken—but an unreliable *detector* (change-detection conflates cosmetic rewrites with flaw detection). Cost: the symbolic path decides in $\sim 0.2 \mu s$ locally with zero API calls; the hybrid adds $k \times \text{steps}$ extraction calls ($\sim 12\text{--}24$, batchable); each judge call generates up to 2,048 tokens. A verifier that is free at decision time changes how often one can afford to check.

5.7 Generality: the Tool-Use Domain

The Tools domain models a travel-booking agent: eight tools, auth scopes as predicates, prerequisite calls threaded as boolean return-value facts (`search_flights` adds `flight-options(trip)`, which `book_flight` requires), an API quota ($-1/\text{call}$) and a spend budget (`flight 300`, `hotel 200`) as resource dimensions. **No changes to any pipeline code were required**—the same labeler, verifier, extractor, and trust model ran as-is, and the domain’s verdicts cross-check 120/120 against the oracle. The one representational wrinkle: the DSL lacks constant action arguments, so scopes are pinned by identity predicates. Notably, the pipeline’s only false rejects arise here, from hyphenated tool-argument extraction—evidence that tool-style vocabularies stress the $NL \rightarrow \text{schema}$ seam more than classical planning vocabularies do.

6 Limitations

All numbers are dev-scale: 360 short-horizon records (72-record test split, 13 negative trust labels) plus 12 records per

domain per long horizon at $H \in \{40, 80\}$ (wide error bars there), one model family (claude-haiku-4-5) for every LLM role, single seed. The long-horizon gold plans are valid but not optimal (Section 4.2), so “horizon H ” is a nominal band, not an optimality claim, and the downstream soundness-gap result (Section 5.2) is demonstrated only at $H \leq 20$, not yet at $H=40, 80$ where the recall trend suggests it would be larger. The trust layer’s value proposition is confined to the prose regime by construction (Section 4.5); the paraphrase step can itself add label noise (all systems see identical text, so comparisons stay like-for-like). The oracle and verifier share design semantics though not code, so a shared semantic error would evade the cross-check; hand-verified fixtures and (future) VAL (Howey, Long, and Fox 2004) validation are the defenses. Precondition similarity is lexical; the DSL omits conditional effects and typed return values; the downstream replan arm is gated by the sound rule-based check, so its “accepted implies valid” property is by construction and the informative quantities are replan success and ungated flawed-execution cost. A remaining open question worth naming plainly: the near-100% extraction-fidelity rate at $H \geq 10$ (Section 5.2) measures agreement between two extraction paths on mechanically repetitive long-horizon plans, not ground-truth extraction accuracy, and should not be read as translation becoming more reliable with plan length.

7 Conclusion

We built and honestly evaluated a hybrid pre-execution verifier for free-text LLM plans. At short horizons (3–8 steps, the range typical of prior LLM-planning benchmarks) its detection matches a sound symbolic check and is beaten to ceiling by a well-budgeted LLM judge—but even there it wins where execution happens: the highest downstream task success (93.1%) and replan feedback that turns rejections into successes more often. That short-horizon parity, however, does not survive scale: at 10–80 steps the hybrid remains perfect while judge recall degrades monotonically, and by 10–20 steps the judge-gated downstream arm already lets flawed plans execute at a nonzero rate the hybrid never does. Four findings reframe the design space: sound verification is a length-invariant guarantee that LLM judging is not, and the gap opens exactly where plans get harder and more error-prone; constrained formats make translation lossless (making the trust problem vanish where they are available); trust signals should flow *both* ways through the fusion rule, triggering re-extraction rather than only extra rejection; and fail-open judge deployments can silently become no verifier at all. The obvious next step is to push the downstream soundness-gap measurement to 40–80 steps and to real (non-synthetic) long-horizon planning tasks, where we expect the gap this laptop-scale infrastructure surfaced to widen further.

Reproducibility

All code, domain generators, prompts, and evaluation scripts are in the project repository; `make reproduce` regenerates every number and figure in this paper from fixed seeds (`make reproduce FULL=1` for the full-scale configu-

ration). Datasets are synthesized locally; the only external dependency is an Anthropic API key.

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